DISPATCH UML Class Diagram

Main

+ main(args : String[]) : void

LoginPage

~frame : JFrame
~layout : CardLayout
~Panel : JPanel
~login : JPanel
~signup : JPanel
~tRegister : JLabel
~forgotPass : JLabel
~typeId : JTextField
~typePass : JPasswordField
~pLogin : JButton
~backToLogin : JLabel
~FName : JTextField
~LName : JTextField
~MName : JTextField
~IDInput : JTextField
~pass : JPasswordField
~conPass : JPasswordField
~bRegister : JButton

+LoginPage()
+showLogin(frame : JFrame) : void
+CreateLoginPanel() : void
+setTextPassFieldCondition(userID : JTextField, userPass : JPasswordField) : void
+login(a : Account, typed : String) : void
+openUserDisplay(role : String, id : int) : void
+realText(field: JTextField, text : String) : String
+RegisterUser() : void
+setFont(text : JLabel) : void
+setFont(text : JLabel, text2 : JTextField) : void
+setFont(text : JLabel, text2 : JLabel) : void
+setFont(text : JLabel, field : JPasswordField) : void
+setMediumFont(text : JLabel) : void
+CreateRegisterPanel() : void
+addPanelComponents(panel : JPanel, components : JComponents...) : void
+displayForgotPassword() : JPanel
+setButton(button : JButton) : void
+actionPerformed(e : ActionEvent) : void
+mouseClicked(e : MouseEvent) : void
+mousePressed(e : MouseEvent) : void
+mouseReleased(e : MouseEvent) : void
+mouseReleased(e : MouseEvent0 : void
+mouseEntered(e : MouseEvent) : void
+mouseExited(e : MouseEvent) : void
+focusGained(e : FocusEvent) : void
+focusLost(e : FocusEvent) : void
+keyTyped(e : KeyEvent) : void
+keyPressed(e : KeyEvent) : void
+keyReleased(e : KeyEvent) : void

JPanel

<<interface>>
ActionListener

+actionPerformed(e : ActionEvent) : void

<<interface>>
MouseListener

+mouseClicked(e : MouseEvent) : void
+mousePressed(e : MouseEvent) : void
+mouseReleased(e : MouseEvent) : void
+mouseReleased(e : MouseEvent0 : void
+mouseEntered(e : MouseEvent) : void
+mouseExited(e : MouseEvent) : void

<<interface>>
FocusListener

+focusGained(e : FocusEvent) : void
+focusLost(e : FocusEvent) : void

<<interface>>
KeyListener

+keyTyped(e : KeyEvent) : void
+keyPressed(e : KeyEvent) : void
+keyReleased(e : KeyEvent) : void

<<interface>>
showLoginPage

+showLogin(frame : JFrame) : void

<<interface>>
SavesLoad

+toLine() : String
+toRow () : Object[]

<<interface>>
Account

+getFullName(): String
+getIdNumber(): int
+getRole(): String
+hasAccount(): boolean
+isBlocked(): boolean
+passwordMatches(typed: String): boolean
+isMustChange(): boolean
+setPassword(newPass: String, mustChange: boolean): void
+recordFailedLogin(): void
+resetTries(): void
+getFailedTries(): int
+save(): void
+clearAccount(): void
+remove(): void

User

idNo: int
lName: String = ""
fName: String = ""
mName: String = ""
password: String = ""
failedTries: int = 0
mustChange: boolean = false

+getIdNumber() : int
+passwordMatches(typed : String) : boolean
+getFailedTries() : int
+isMustChange() : boolean
+getLastName() : String
+getFirstName() : String
+getMiddleName() : String
+getFullName() : String
+hasAccount() : boolean
+setPassword(newPass : String, mustChange : boolean) : void
+setNames(last : String, first : String, middle : String) : void
+recordFailedLogin() : void
+resetTries() : void
+isBlocked() : boolean
+clearAccount() : void

UserDisplay

~SEARCH_TEXT : String = "Search by name, ID or serial no..."
~t1: JLabel
~tRole : JLabel
~logout: JLabel
~line : JLabel
~line2: JLabel
~h1: JLabel
~h2: JLabel
~h3: JLabel
~mediumFont : Font
~headerFont : Font
~browsePane : JPanel
~scrollBrowsePane : JScrollPane
~requestPane : JPanel
~profile : ProfilePanel
~scrollProfile : JScrollPane
~lastRequestFileTime : long = -1
~editProfile: JButton
~requestReset: JButton
~requestDelete: JButton
~noticeOpen : boolean = false
~search : JTextField
~layout : CardLayout
~set : JPanel
~currentStudent : Student = null
~currentTeacher : Teacher = null
~browseBoxHolder : JPanel
~browseOverlay : JPanel
~browseDetailPanel : JPanel
~image : JLabel
~name : JLabel
~desc : JLabel
~qty : JLabel
~statusT : JLabel
~accessT : JLabel
~addRequest : JButton
-quantity : int = 1
~addQuantity : JButton
~minusQuantity : JButton
~quantityField : JTextField
~selectedItem : AdminDisplay.Equipment
~requestTable : JTable
~requestDtm : DefaultTableModel
~pFull : JLabel
~txtID : JLabel
~txtRole : JLabel
~txtLName : JLabel
~txtFName : JLabel
~txtMName : JLabel

+UserDisplay()
+main(args : String[]) : void
+minQuantityPerEquipment() : int
+maxQuantityPerEquipment() : int
+getMostAllowed() : int
+setrequestPaneDisplay(panel : JPanel) : void
+showSelectedReceipt() : void
+setupBrowseSearch(panel : JPanel) : void
+refreshMyRequests() : void
+setrbrowsePaneDisplay(panel : JPanel) : void
+loadBrowseEquipment() : void
+showBrowseEquipment() : void
+createEquipmentBrowse(box : JLabel, x : int, y : int, width : int, height : int, item : AdminDisplay.Equipment)
+setupBrowseOverlay() : void
+showBrowseOverlay(item : AdminDisplay.Equipment) : void
+hideBrowseOverlay() : void
+setImageButton(button : JButton) : void
+bufferImage(img : JLabel, icon : ImageIcon) : void
+setprofilePaneDisplay(panel : JPanel) : void
+editProfile() : void
+changePassword() : void
+loadProfile() : void
+showProfile(roleText : String, id : int, last : String, first : String, middle : String) : void
+setProfileFontPlain(text : JLabel, x : int, y : int, width : int, height : int) : void
+setProfileFontBold(text : JLabel, x : int, y : int, width : int, height : int) : void
+setButtonProfile(button: JButton, x : int, y : int, width : int, height : int) : void
+setupProfileButtons() : void
+setHeaderFont(text : JLabel, color : String) : void
+showApprovedNotice() : void
+setMediumFont(text :JLabel) : void
+setMediumHeader(text :JLabel) : void
+getQuantityNumber() : int
+getUserId() : int
+getUserRole() : String
+startNoticeWatcher() : void
+actionPerformed(e : ActionEvent) : void
+mouseClicked(e : MouseEvent) : void
+mousePressed(e : MouseEvent) : void
+mouseReleased(e : MouseEvent) : void
+mouseReleased(e : MouseEvent0 : void
+mouseEntered(e : MouseEvent) : void
+mouseExited(e : MouseEvent) : void
+focusGained(e : FocusEvent) : void
+focusLost(e : FocusEvent) : void
+keyTyped(e : KeyEvent) : void
+keyPressed(e : KeyEvent) : void
+keyReleased(e : KeyEvent) : void

Student

~role: String = "Student"
~studentList: ArrayList<Student>
~Scolumns: String[] = {"Student ID","Photo","Last Name","First Name","Middle Name","Gender","Course"...}
~status: String[] = {"Active","Inactive"}
-course: String = ""
-year: String = ""
-borrowed: String = ""
-studentStatus: String = ""

+setDetails(idNO : int, status : String) : void
+addStudentToFile(idNo : String, status : String) : Student
+saveStudentFile() : void
+catchLetter(input : String) : boolean
+findIDNumber(id : int) : Student
+loadStudentInfo() : void
+setSavedStudentInfo(idNo : int, lName : String, fName : String, mName : String, department : String, borrowed : String, studentStatus : String, password : String) : void
+refreshStudentTable(studentDtm : DefaultTableModel) : void
+registerAccount(id : String, Lname : String, Fname : String, Mname : String, pass : String, confirmPass : String) : boolean
+unBlock(id : int) : boolean
+changeStatus(id : int, newStatus : String) : boolean
+updateProfile(id : int, newLast : String, newFIrst : String, newMiddle : String, newCourse : String, newYear : String) : boolean
+changePassword(id : int, oldPass : String, newPass : String) : String
+getStudentCourse() : String
+getBorrowed() : String
+getYear() : String
+getStudentStatus() : String
+toLine() : String
+toRow() : Object[]
+getRole() : String
+save() : void
+remove() : void

Teacher

~role: String = "Teacher"
~teacherList: ArrayList<Teacher>
~Tcolumns: String[] = {"Teacher ID","Photo","Last Name","First Name","Middle Name","Gender","Department"...}
~status: String[] = {"Active","Inactive"}
-department: String = ""
-borrowed: String = ""
-teacherStatus: String = ""

+setDetails(idNO : int, status : String) : void
+addTeacherToFile(idText : String, status : String) : Teacher
+saveTeacherFile() : void
+catchLetter(input : String) : boolean
+findIDNumber(id : int) : Teacher
+loadTeacherInfo() : void
+setSavedTeacherInfo(idNo : int, lName : String, fName : String, mName : String, department : String, borrowed : String, status : String, password : String) : void
+refreshTeacherTable(teacherDtm : DefaultTableModel) : void
+registerAccount(id : String, Lname : String, Fname : String, Mname : String, pass : String, conPass : String) : boolean
+unBlock(id : int) : boolean
+changeStatus(id : int, newStatus : String) : boolean
+updateProfile(id : int, newLast : String, newFIrst : String, newMiddle : String, newDepartment : String) : boolean
+changePassword(id : int, oldPass : String, newPass : String) : String
+getDepartment() : String
+getBorrowed() : String
+getTeacherStatus() : String
+toLine() : String
+toRow() : Object[]
+getRole() : String
+save() : void
+remove() : void

BlockedUsers

~blockUserscolumns: String[] = {"User","Role","Reason","Date","Status"}
~MAX_TRIES: int = 3
~blockUsers: ArrayList<BlockedUsers>
-user: String
-role: String
-reason: String
-date: String
-status: String

+BlockedUsers(user: String, role: String, reason: String, date: String, status: String)
+getUser(): String
+getRole(): String
+getReason(): String
+getDate(): String
+getStatus(): String
+find(id: String, role: String): BlockedUsers
+getBlockedList(): ArrayList<BlockedUsers>
+saveBlockedUsers(): void
+loadBlockedUsers(): void
+loadBlockUsersTable(dtm: DefaultTableModel): void
+block(id: String, role: String, reason: String): void
+isBlocked(user: String, role: String): boolean
+isBlockedNow(user: String, role: String): Boolean
+unblock(id: String, role: String): boolean
+toLine(): String
+toRow(): Object[]

Admin

~role: String = "Admin"
~ADMIN_ID: int = 2026001022
-DEFAULT_PASSWORD: String = "asd1234acjn"
-instance: Admin

-Admin()
+getInstance(): Admin
+isAdminId(id: int): boolean
+load(): void
+save(): void
+getRole(): String
+isBlocked(): boolean
+recordFailedLogin(): void
+resetTries(): void
+clearAccount(): void
+remove(): void

ProfilePanel

~GENDER_FILE: String
~GENDER_NONE: String
+pFull: JLabel
+txtID: JLabel
+txtRole: JLabel
+txtFName: JLabel
+txtMName: JLabel
+txtLName: JLabel
+txtGender: JLabel
+titleA: JLabel
+valueA: JLabel
+titleB: JLabel
+valueB: JLabel
+picture: JLabel
+editProfile: JButton
+editPassword: JButton
+infoPane: JPanel

+ProfilePanel()
-addField(title: String, value: JLabel, centerX: int, y: int): void
-addField(label: JLabel, value: JLabel, centerX: int, y: int): void
+addButton(button: JButton, x: int): void
+show(role: String, id: int, last: String, first: String, middle: String): void
+show(role: String, id: int, last: String, first: String, middle: String, tA: String, a: String, tB: String, b: String): void
+showExtras(tA: String, a: String, tB: String, b: String): void
+getGender(role: String, id: int): String
+setGender(role: String, id: int, gender: String): void
+deleteGender(role: String, id: int): void
-otherUsersGender(role: String, id: int): ArrayList<String>
+askNames(parent: Component, role: String, id: int, last: String, first: String, middle: String): String[]
+askNames(parent: Component, role: String, id: int, last: String, first: String, middle: String, titleA: String, valueA: String, titleB: String, valueB: String): String[]

ProfilePic

~TYPES: String[] = {"png", "jpg", "jpeg"}
~MAX_BYTES: long = 5L * 1024 * 1024
~cache: HashMap<String, ImageIcon>

-name(role: String, id: int): String
+find(role: String, id: int): String
+browse(parent: Component): String
+save(role: String, id: int, sourcePath: String): void
+delete(role: String, id: int): void
+load(role: String, id: int, w: int, h: int): ImageIcon
+fromFile(path: String, w: int, h: int): ImageIcon

PasswordReset

~ADMIN_OFFICE: String = "the Admin Office, Center Building"
~CODE_CHARS: String = "ABCDEFGHJKLMNPQRSTUVWXYZ23456789"
~CODE_LENGTH: int = 8
~random: SecureRandom

+generateCode(): String
+findAccount(id: int, role: String): Account
+checkNewPassword(pass: String, confirm: String): String
+showRequestInstructions(parent: Component): void
+adminReset(parent: Component, id: int, role: String): void
+forceChange(parent: Component, id: int, role: String): boolean

ManageFiles

~BASE: String = "C:\\Users\\ADMIN\\OneDrive\\Documents\\Project OOP\\"
~EQUIPMENT: String = BASE + "Admin\\Equipments\\equipment.txt"
~REQUEST: String = BASE + "Admin\\Requests\\requests.txt"
~RULES: String = BASE + "Admin\\Requests\\rules.txt"
~STUDENT: String = BASE + "Admin\\Registered Users\\Student.txt"
~TEACHER: String = BASE + "Admin\\Registered Users\\Teacher.txt"
~BLOCKED: String = BASE + "Admin\\Blocked Users\\BlockedUsers.txt"
~IMAGE_FOLDER: String = BASE + "Equipment Images"
~PROOF_FOLDER: String = BASE + "Request Proofs"
~PICTURE_FOLDER: String = BASE + "Profile Pictures"
~ADMIN: String = BASE + "Admin\\Admin.txt"
~DELETE: String = BASE + "Admin\\Registered Users\\DeleteRequests.txt"

+clean(text: String): String
+join(fields: Object[]): String
+saveFile(path: String, list: ArrayList<? extends SavesLoad>): void
+writeLines(path: String, lines: ArrayList<String>): void
+loadFile(path: String): ArrayList<String[]>
+fillTable(dtm: DefaultTableModel, list: ArrayList<? extends SavesLoad>): void
+copyFile(sourcePath: String, folderPath: String, newName: String): String
+loadScaledIcon(path: String, maxW: int, maxH: int): ImageIcon

DeleteRequests

~columns: String[*] = {"User ID","Role","Reason","Date","Status"}
~list: ArrayList<DeleteRequests>
-userId: int
-role: String-reason: String
-date: String
-status: String

+toLine(): String
+toRow(): Object[]
+DeleteRequests(userId: int, role: String, reason: String, date: String, status: String)
+getUserId(): int
+getRole(): String
+getReason(): String
+getDate(): String
+getStatus(): String
+save(): void
+load(): void
+findPending(id: int, role: String): DeleteRequests
add(id: int, role: String, reason: String): void
+setStatus(newStatus: String): void
loadTable(dtm: DefaultTableModel): void

Receipt

+buildReceipt(r: AdminDisplay.Request): String
-chargeLines(r: AdminDisplay.Request): String
+show(parent: Component, r: AdminDisplay.Request): void

Receipt

+checkCanDelete(id: int, role: String): String
+removeAccount(id: int, role: String, removeId: boolean): void
+adminDelete(parent: Component, id: int, role: String): boolean

RequestQueue

~queue: int[]
~front: int = 0
~rear: int = -1

+isEmpty(): boolean
+isFull(): boolean
+size(): int
+clear(): void
+enqueue(data: int): void
+peek(): int
+dequeue(): int

SearchHelper

+binarySearch(arr: int[], low: int, high: int, target: int): int
+firstMatch(keys: String[], low: int, high: int, target: String): int
+insertionSort(array: int[], n: int, unsorted: int[]): void
+insertionSort<T>(list: ArrayList<T>, order: Comparator<T>): void

AdminDisplay

-private adminAccount: int = 2026001022
-private password: String = "asd1234acjn"
~adminLast: String = "Admin"
~adminFirst: String = "System"
~adminMiddle: String = ""
~name: String = "Admin"
+user: JLabel
+logout: JLabel
+layout: CardLayout
+mainPane: JPanel
+dashboardPane: JPanel
+equipmentPane: JPanel
+requestPane: JPanel
+systemPane: JPanel
+adminProfile: ProfilePanel
+line1: JLabel
+line2: JLabel
+search: JTextField
+h1: JLabel
+h2: JLabel
+h3: JLabel
+h4: JLabel
+h5: JLabel
+h6: JLabel
+lastRequestFileTime: long = -1
+hStudent: JLabel
+hTeacher: JLabel
+hBlockUsers: JLabel
+pane: JPanel
+userCard: JPanel
+userPanels: CardLayout
+equipmentTable: JTable
+studentTable: JTable
+teacherTable: JTable
+blockUserTable: JTable
+categories: JComboBox
+dtm: DefaultTableModel
+studentDtm: DefaultTableModel
+teacherDtm: DefaultTableModel
+blockedUserDtm: DefaultTableModel
+recentTable: JTable
+recentDtm: DefaultTableModel
+lastRequestCount: int = -1
~RECENT_LIMIT: int = 10
+selectedCategory: String
+selectedStatus: String
+panelAdd: JPanel
+panelAddT: JPanel
-private proofViewPanel: JPanel
+proofFull: JLabel
+currentProofPath: String = ""
+teacherID: JTextField
+studentID: JTextField
+inputStudentPer: JTextField
+inputTeacherPer: JTextField
+inputPickup: JTextField
+statusBox: JComboBox<String>
+requestFilter: JComboBox<String>
+accessBox: JComboBox<String>
+categoryBox: JComboBox<String>
+status: JComboBox<String>
+statusBoxT: JComboBox<String>
+seNoField: JTextField
+qtyField: JTextField
+eqNameField: JTextField
+searchStudent: JTextField
+searchTeacher: JTextField
+searchBlocked: JTextField
+locationField: JTextField
+descriptionArea: JTextArea
+detailArea: JTextArea
+reasonField: JTextArea
+imageName: JLabel
+selectedImagePath: String = ""
+changeImage: JButton
+editImagePath: String
+box1: JLabel
+box2: JLabel
+box3: JLabel
+box4: JLabel
+box5: JLabel
+proofPicture: JLabel
+requestTable: JTable
+requestDtm: DefaultTableModel
+inputMaxStudent: JTextField
+inputMaxTeacher: JTextField
+inputMaxDays: JTextField
+inputPenalty: JTextField
+resetStudent: JButton
+editStudent: JButton
+editTeacher: JButton
+resetTeacher: JButton
+deleteStudent: JButton
+rbutton: JButton
+deleteTeacher: JButton
+approveDelete: JButton
+rejectDelete: JButton
+hDelete: JLabel
+deleteTable: JTable
+deleteDtm: DefaultTableModel
+lastQueueSize: int = -1
+lastQueueFront: int = -1
-private overlay: JPanel
-private addEquipmentPanel: JPanel

+AdminDisplay()
+mouseClicked(e: MouseEvent): void
+showUserTab(card: String, tab: JLabel): void
+showTab(card: String, tab: JLabel): void
+getAdminPassword(): String
+highlight(selected: JLabel, group: JLabel...): void
+mousePressed(e: MouseEvent): void
+mouseReleased(e: MouseEvent): void
+mouseEntered(e: MouseEvent): void
+mouseExited(e: MouseEvent): void
+actionPerformed(e: ActionEvent): void
+setlocations(l1: JLabel, l2: JLabel, h1: JLabel, h2: JLabel, h3: JLabel, h4: JLabel, h5: JLabel, h6: JLabel): void
+refreshDashboard(): void
+setDashboardPane(dashboard: JPanel): void
+setEquipmentPane(equipmentPane: JPanel): void
+setRequestPane(requestPane: JPanel): void
+releaseRequest(): void
+blockRequestUser(): void
+deleteSelectedRequest(): void
+loadRequestTable(): void
+startQueueWatcher(): void
+makeTable(holder: JPanel, columns: String[], rowHeight: int, width: int, height: int): JTable
+getTargetRequest(): Request
+showRequestDetails(): void
+loadQueueRequestTable(): void
+getSelectedRequest(): Request
+loadProofIcon(path: String, maxW: int, maxH: int): ImageIcon
+approveRequest(): void
+resetUserPassword(table: JTable, model: DefaultTableModel, role: String): void
+deleteUserAccount(table: JTable, model: DefaultTableModel, role: String): void
+rejectRequest(): void
+applyRules(): void
+setSidePanelButtons(button: JButton, x: int, y: int, width: int): void
+getSelectedDeleteRequest(): DeleteRequests
+approveDeleteRequest(): void
+rejectDeleteRequest(): void
+setSystemPane(systemPane: JPanel): void
+loadAdmin(): void
+saveAdmin(): void
+checkAdminPassword(typed: String): boolean
+returnRequest(): void
+setEquipmentTables(holder: JPanel): void
+setRegisteredStudentID(holder: JPanel, columns: String[]): void
+setRegisteredTeacherID(holder: JPanel, columns: String[]): void
+fixPhotoColumn(table: JTable): void
+setBlockedUsers(holder: JPanel, columns: String[]): void
+setButtons(button: JButton, x: int, y: int, width: int, height: int): void
+setRecentActivityTables(pane: JPanel): JPanel
+loadRecentActivity(): void
+setupAdminProfile(): void
+loadAdminProfile(): void
+editAdminProfile(): void
+changeAdminPassword(): void
+getBorrowerName(r: Request): String
+setupAddEquipmentOverlay(): void
+setupProofOverlay(): void
+showProofOverlay(): void
+showOverlay(panel: JPanel): void
+hideOverlay(): void
+chooseEquipmentImage(): void
+BrowseImage(): String
+getEquipmentIcon(path: String): ImageIcon
+loadEquipmentTable(): void
+deleteEquipment(): void
+editEquipment(): void
+editStudentStatus(): void
+setJTextArea(holder: JScrollPane, title: String): void
+setTitledBorderTextfield(field: JTextField, title: String): void
+editTeacherStatus(): void
+unblockUser(): void
+setTitledBorderTextfield(field: JPanel, title: String): void
+setComboBox(comboBox: JComboBox): void
+showAddEquipment(): void
+addEquipment(e: Equipment): void
+hideAddEquipment(): void
+getKeyword(field: JTextField, hint: String): String
+getSelectedUserId(table: JTable, model: DefaultTableModel, role: String): int
+refreshUserTables(): void
+confirmAdminPassword(): boolean
+focusGained(e: FocusEvent): void
+focusLost(e: FocusEvent): void
+keyTyped(e: KeyEvent): void
+keyPressed(e: KeyEvent): void
+keyReleased(e: KeyEvent): void
+searchEquipment(): void
+searchStudents(): void
+searchTeachers(): void
+searchBlockedUsers(): void